

Erqiang Tang

Ph.D. Candidate, Electronic and Computer Engineering (ECE), HKUST

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Research Interests

My research lies at the intersection of generative modeling and wireless communications, with a focus on harnessing modern generative AI techniques for wireless signal processing.

Education

The Hong Kong University of Science and Technology (HKUST)

Ph.D. in Electronic and Computer Engineering

2022 – Present

Advisor: Prof. Khaled B. Letaief

The Hong Kong University of Science and Technology (HKUST)

B.Eng. in Electronic Engineering, Minor in Design

2018 – 2022

CGA: 3.98 / 4.30

Publications

* Equal contribution (co-first authors).

1. T. Liao*, **E. Tang***, Z. Wang, P. Biswas, J. Sun, W. Guo, J. Qian, S. Song, J. Zhang, K. B. Letaief. "Scaling Fluid Antenna Systems: Generative Learning and Decentralized Optimization for Efficient Air-Interface Design." *Under review*.
2. **E. Tang**, W. Guo, H. He, S. Song, J. Zhang, K. B. Letaief. "A Unified Two-Stage Generative Diffusion Framework for Channel Estimation and Port Selection in Multiuser MIMO-FAS." *Under review*.
3. **E. Tang**, W. Guo, H. He, S. Song, J. Zhang, K. B. Letaief. "Accurate and Fast Channel Estimation for Fluid Antenna Systems with Diffusion Models." *IEEE Global Communications Conference (GLOBECOM)*, 2025, pp. 2066–2071.

Teaching Experience

Teaching Assistant — Department of Electronic and Computer Engineering, HKUST

2022 – 2023

- EESM 5539 — Wireless Communication Networks
- EESM 5670 — Advanced Architectures and Designs for Communication Networks
- ELEC 3300 — Introduction to Embedded Systems
- ELEC 3100 — Signal Processing and Communications

Delivered tutorial sessions, graded homework and exams, and mentored undergraduate final projects.

Honors & Awards

HKUST Postgraduate Studentship (PGS)

2022 – Present

HKUST Academic Achievement Medal (top 1% of graduates)

2022

HKUST School of Engineering Dean's List

2018 – 2021

Professional Service

Journal Reviewer: IEEE Wireless Communications Letters

Conference Reviewer: IEEE Global Communications Conference (GLOBECOM)

Technical Skills

MATLAB, Python (PyTorch), LaTeX

Languages

Mandarin (Native) English (Professional) Cantonese (Conversational)